

Swine Pudendal Nerve Intraneural Stimulation Enables Selective Neuromodulation of Urethral and Anal External Sphincters

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Abstract—Precise neuromodulation of the pudendal nerve represents a promising approach to restore micturition control in individuals affected by lower urinary tract dysfunctions. However, currently approved electroceuticals suffer from limited stimulation selectivity, often leading to suboptimal outcomes due to off-target activation of neural pathways. To address this limitation, the use of Transverse Intrafascicular Multichannel Electrode (TIME) was investigated in a swine model, to achieve selective control of the external anal and urethral sphincters. We compared two implantation sites along the pudendal nerve via transgluteal surgical access: a proximal location in the sacral region before trunk consolidation, and a more distal site at the anatomical convergence of sacral contributions. Proximal stimulation selectively recruited only the anal sphincter, while distal placement enabled selective activation of both muscles. These findings highlight the importance of implant positioning and support the use of TIME as a powerful solution for precise neuroprosthetic control of urinary functions.

I. INTRODUCTION

LOWER urinary tract dysfunctions, causing incontinence or urinary retention, severely impact life quality of affected individuals. Current neuromodulation approaches like sacral nerve stimulation and posterior tibial nerve stimulation use lead and needle electrodes, which lack selectivity - i.e. the capability to precisely stimulate portions of the nerve - often causing undesired effects and limiting therapeutical efficacy [1]. The pudendal nerve is a promising alternative due to its dual role: carrying afferent fibers encoding bladder fullness and efferent fibers innervating the anal and urethral external sphincters, thus enabling direct modulation of continence mechanisms. In this study, TIME was investigated as a selective solution for neuroprosthetic control of urinary functions.

II. METHODS

The protocol for all the animal studies (No. 278/2021-PR) was approved by the Italian Ministry of Health and was in accordance with the Italian law (D.lgs. 26/2014)

Two experiments were conducted to evaluate muscle-specific responses to pudendal nerve stimulation by recording the electromyographic (EMG) activity from the external urethral sphincter (EUS) and the external anal sphincter (EAS). Procedures included two male farm pigs (*Sus Scrofa Domestica*) weighing 35-40 kg, aged 3-4 months. The nerve was surgically exposed as described in [2], and a TIME was implanted. In Experiment 1 the electrode was placed in the sacral region of the nerve before it consolidates into a distinct trunk, while in Experiment 2 it was implanted more distally, at the anatomical convergence of pudendal sacral contributions (Figure 1: a). Each electrode featured 16 active sites (ASs), 60 μm in diameter and spaced 134 μm (Figure 1: b). Biphasic, cathodic-first pulses (200 μs pulse width) were delivered with increasing current amplitudes from 10 to 600 μA . This range was divided into 50 equally spaced steps and three repetitions

per amplitude were performed producing pulses with a frequency of 3 Hz. EMG signals from the EUS and EAS were recorded using needle electrodes. Recruitment curves for each muscle and AS ($\overline{EUS}$ and $\overline{EAS}$) were obtained as in [3] and their maximum difference was used to quantify selectivity:

$$\text{Selectivity} = \max(\overline{EUS} - \overline{EAS}).$$

While not the standard metric, this proves particularly useful in binary comparisons, offering an easily interpretable overview of all active sites on the electrode.

III. RESULTS & DISCUSSION

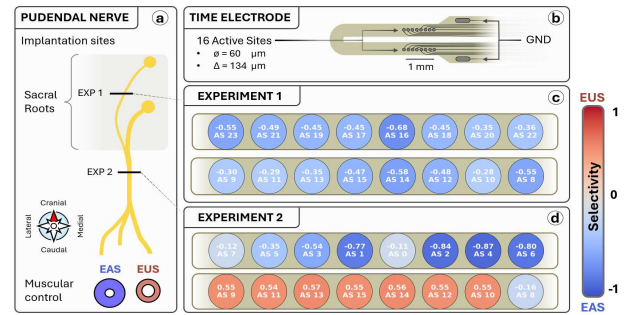


Figure 1: (a) Pudendal nerve and implantation sites. (b) TIME electrode. Results from Experiments 1 (c) and 2 (d).

Experiment 1 showed consistent EAS recruitment across all ASs, with selectivity values from -0.68 to -0.28 (Figure 1: c). This suggests a distribution of fibers favoring EAS recruitment when stimulating at this level of the pudendal nerve. In contrast, Experiment 2, where TIME was placed more caudally, revealed a clear shift in selectivity: seven active sites preferentially activated the EUS (e.g., AS11, AS14), while four were strongly EAS-selective (e.g., AS4; AS6) (Figure 1: d). These findings suggest the presence of a functional fascicular gradient along the pudendal nerve that can be exploited to enhance stimulation selectivity and reduce off-target effects.

The TIME emerges as a powerful solution to the spatial selectivity challenges, enabling targeted modulation of key effectors involved in continence. However, the results underscore the critical role of the implantation site in achieving precise activation. The differences in recruitment patterns observed between proximal and distal placements warrants further anatomical and functional mapping.

Nevertheless, these findings lay the groundwork for more refined and effective neuroprosthetic strategies for lower urinary tract control that exploit the pudendal nerve as implantation site and TIMEs as neural interfaces.

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